

DIFFERENTIAL GENE EXPRESSION ANALYSIS OF BREAST CANCER

Using TCGA-BRCA RNA-seq Data

Study type	Independent computational bioinformatics analysis
Dataset	The Cancer Genome Atlas — Breast Cancer (TCGA-BRCA)
Comparison	Primary Tumor vs Solid Tissue Normal
Platform	R / Bioconductor
Core tools	TCGAbiolinks, DESeq2, EnhancedVolcano, pheatmap, clusterProfiler

Research Question

Which genes are differentially expressed between breast tumour and normal breast tissue in TCGA-BRCA, and what biological processes are associated with these changes?

Project Scope

This report presents a reproducible research-training/portfolio analysis of publicly available TCGA-BRCA bulk RNA-seq data. It is not a clinical diagnostic or treatment study.

1. Abstract

Breast cancer is a molecularly heterogeneous disease in which tumour and normal tissues exhibit distinct transcriptional programs. This project used publicly available TCGA-BRCA RNA-seq count data to identify differentially expressed genes between primary breast tumours and solid tissue normal samples. TCGAbiolinks was used for data retrieval and preparation and DESeq2 was used for differential-expression analysis. Metastatic samples were excluded. The final comparison included 1,111 Primary Tumor samples and 113 Solid Tissue Normal samples. After low-count filtering, 56,705 genes were analysed. With $\text{padj} < 0.05$ and $|\log_2 \text{fold change}| > 1$, 12,304 significant DEGs were identified, comprising 8,186 upregulated and 4,118 downregulated genes. PCA demonstrated broad separation of tumour and normal expression profiles. Gene Ontology enrichment highlighted chromosome organization, nuclear division and cell-cycle-associated processes among upregulated genes, while circulation, muscle, chemotaxis and vascular-development processes were enriched among downregulated genes.

2. Objectives

- Retrieve and preprocess TCGA-BRCA RNA-seq count data.
- Compare primary tumour and normal tissue using DESeq2.
- Identify significant upregulated and downregulated genes.
- Visualize the results using PCA, volcano plots and heatmaps.
- Perform Gene Ontology Biological Process enrichment.
- Interpret findings with appropriate bulk-RNA-seq limitations.

3. Dataset and Sample Selection

The dataset was queried from the Genomic Data Commons for TCGA-BRCA using the STAR - Counts workflow. Primary Tumor and Solid Tissue Normal samples were retained for the stated research question; metastatic samples were excluded.

Measure	Result
Total samples	1,224
Primary Tumor	1,111
Solid Tissue Normal	113
Genes after filtering	56,705
Significant DEGs	12,304
Upregulated	8,186
Downregulated	4,118

4. Materials and Methods

4.1 Count preprocessing

The unstranded count matrix was used as input for DESeq2. Ensembl version suffixes were removed, duplicate identifiers were combined by summing counts, and genes with a total count below 10 were filtered out. This yielded 56,705 genes.

4.2 Differential expression

DESeq2 modelled the count data with tissue condition as the experimental factor. Solid Tissue Normal was the reference and Primary Tumor was the comparison. Multiple testing was controlled using the Benjamini-Hochberg procedure.

4.3 Significance criteria

Genes were considered significant when adjusted p-value was < 0.05 and absolute log2 fold change was > 1 . Positive log2 fold change represents higher expression in primary tumour relative to normal tissue.

4.4 Visualization

Variance-stabilizing transformation was used for sample-level visualization. PCA assessed global expression structure; EnhancedVolcano visualized fold change and statistical significance; heatmap was used for clustered heatmaps.

4.5 Functional enrichment

Gene identifiers were annotated with org.Hs.eg.db. Upregulated and downregulated genes were analysed separately for Gene Ontology Biological Process enrichment using clusterProfiler.

5. Analysis Workflow

TCGA-BRCA RNA-seq	Sample selection Tumour + Normal	Count preprocessing IDs + filtering	DESeq2 DGE analysis	PCA / Volcano / Heatmaps	GO enrichment Interpretation
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6. Results: Global Expression Structure

PCA showed broad separation between primary tumour and normal samples. PC1 explained 19% of the variance and PC2 explained 12%, indicating that tissue condition contributes substantially to the global expression structure.

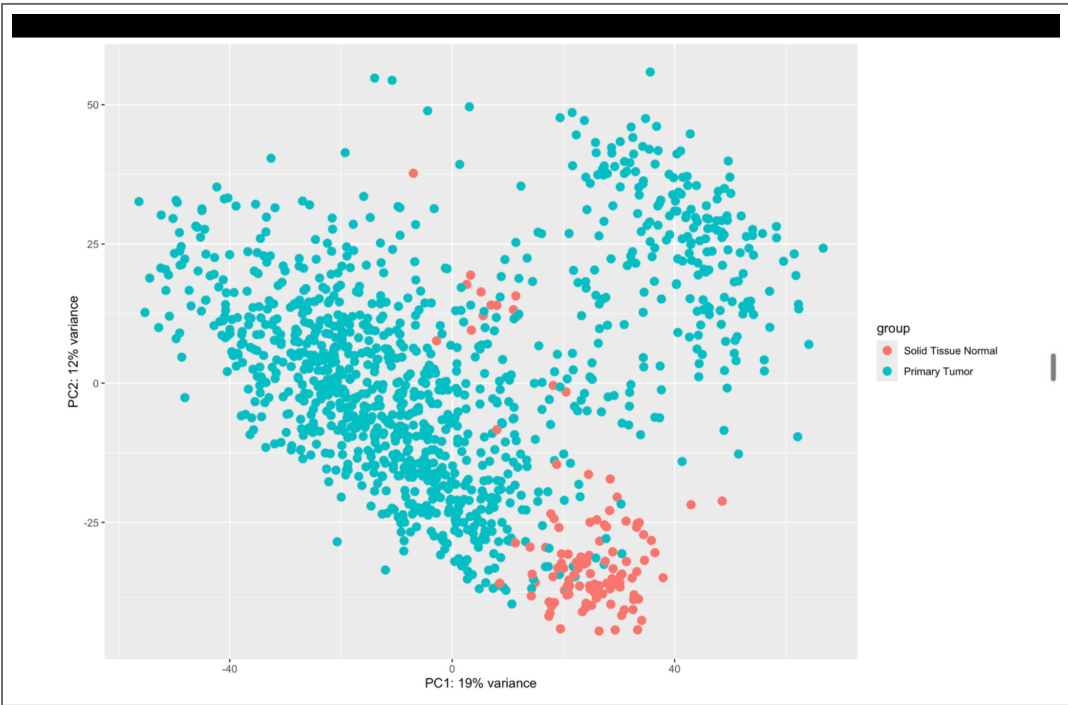


Figure 1. PCA of TCGA-BRCA samples. PC1 = 19% variance; PC2 = 12% variance.

7. Differential Expression: Volcano Plot

The volcano plot displays log2 fold change against $-\log_{10}$ adjusted p-value. The dashed vertical lines mark ± 1 log2 fold change and the horizontal line marks the adjusted p-value threshold. Genes beyond both criteria were classified as significant DEGs.

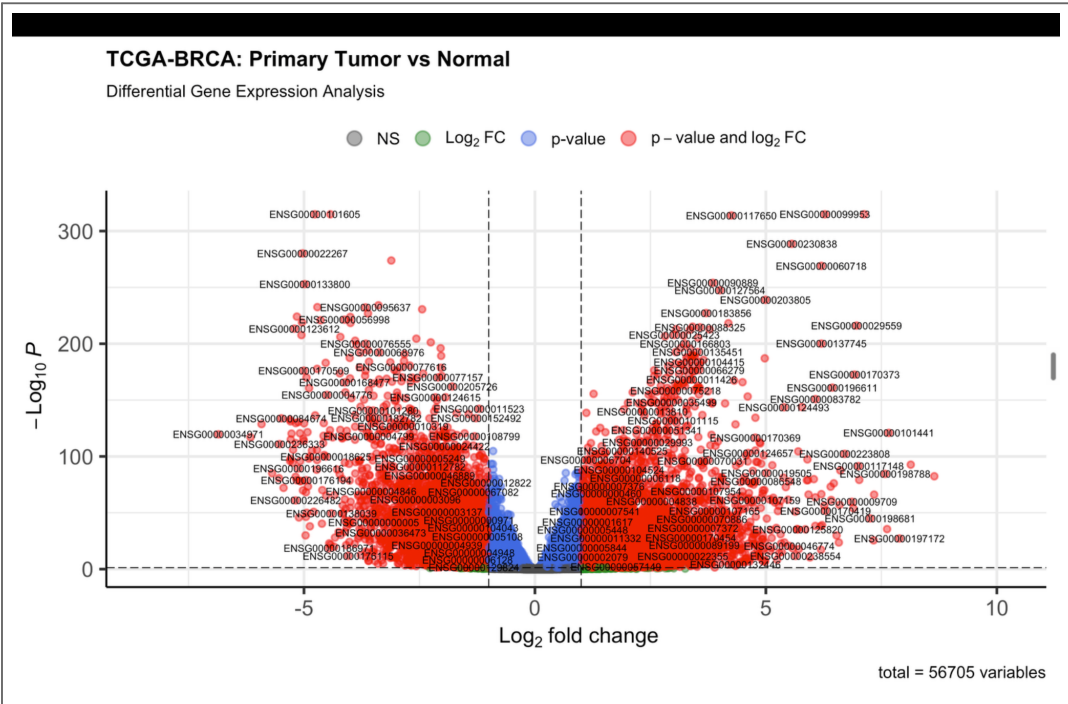


Figure 2. Volcano plot for Primary Tumor versus Solid Tissue Normal.

8. Expression Heatmaps

Hierarchical clustering provides a visual summary of gene- and sample-level expression patterns. The top-50 heatmap gives an overview of the strongest significant signals, while the final top-30 heatmap uses gene symbols and is intended as the cleaner portfolio figure.

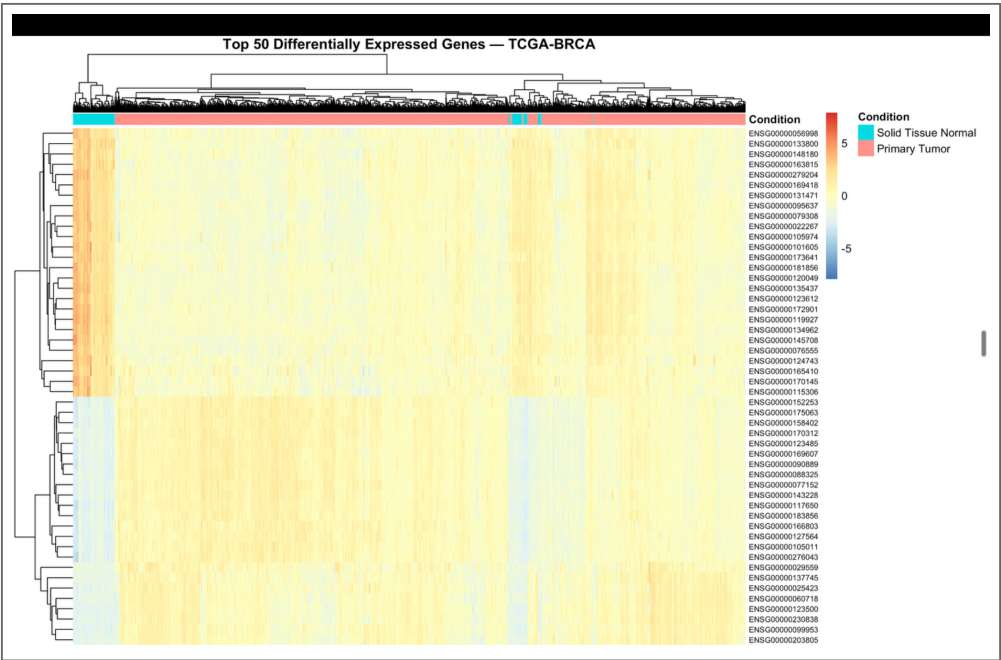


Figure 3. Top-50 differentially expressed genes — TCGA-BRCA.

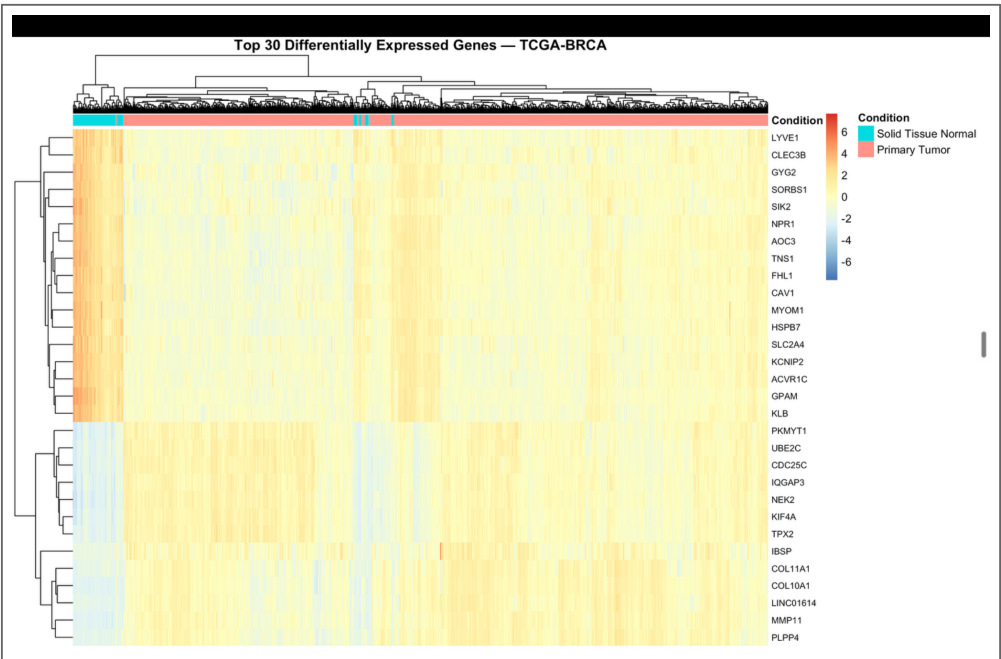


Figure 4. Final top-30 differential-expression heatmap with gene symbols and condition annotation.

9. Selected Breast-Cancer-Associated Genes

BRCA1, BRCA2, ESR1 and ERBB2 met the project's combined statistical and fold-change criteria. TP53 had a significant adjusted p-value but did not meet the $|\log_2FC| > 1$ effect-size threshold.

Gene	log2FC	Adjusted p-value	Project classification
BRCA1	+1.107	1.59×10^{-33}	Significant
BRCA2	+1.570	4.70×10^{-33}	Significant
ERBB2	+1.903	7.97×10^{-31}	Significant
ESR1	+1.721	3.30×10^{-31}	Significant
TP53	+0.227	4.65×10^{-3}	Not a DEG by $ \log_2FC > 1$

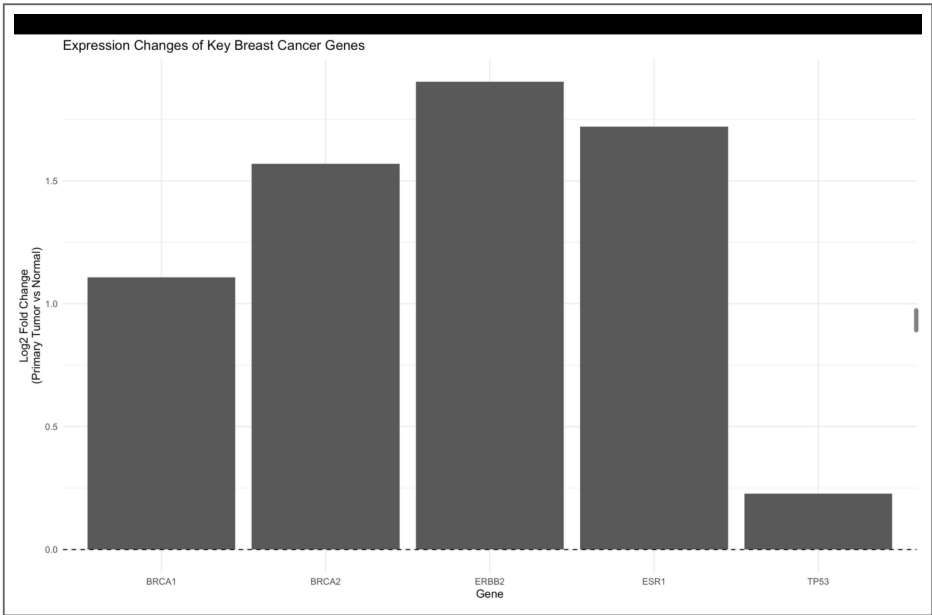


Figure 5. Log2 fold changes of selected breast-cancer-associated genes.

Interpretation: These are bulk RNA-seq expression differences. They do not establish BRCA1/BRCA2 mutation status, protein activity, or clinical causality.

10. Gene Ontology Biological Process Enrichment

Upregulated genes were enriched for organelle fission, nuclear division, chromosome segregation, nuclear chromosome segregation, nucleosome organization and assembly, and cell-cycle checkpoint-related processes. These findings are consistent with increased representation of proliferation and chromosome-maintenance programs in tumour tissue.

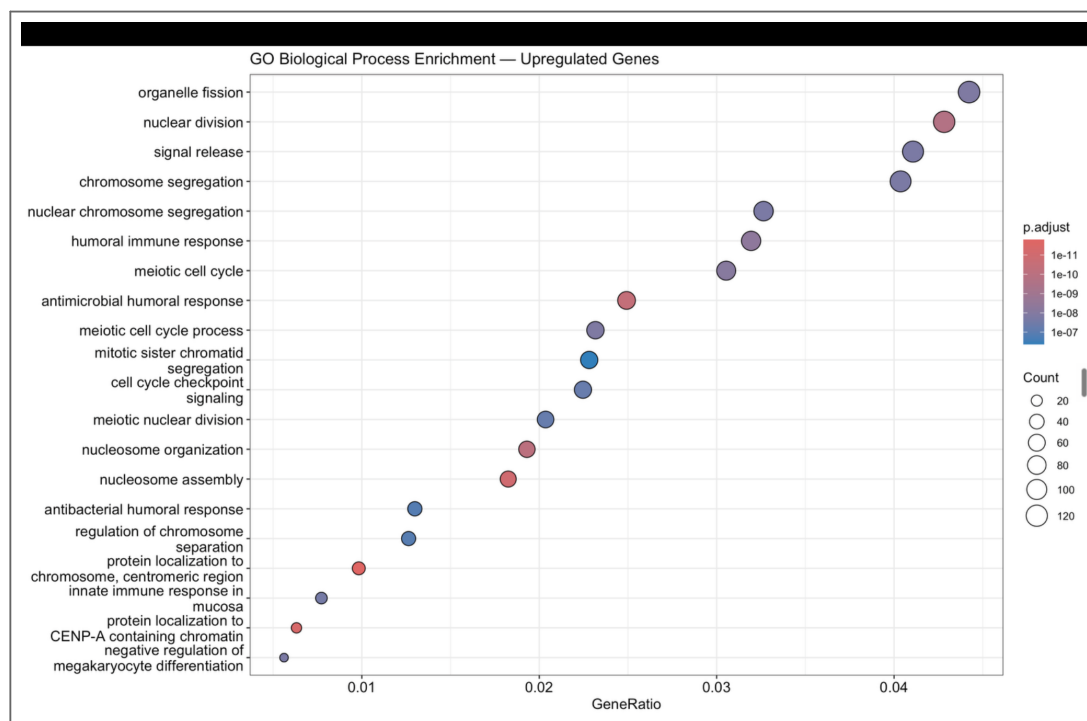


Figure 6. GO Biological Process enrichment of upregulated genes.

Downregulated genes were enriched for blood circulation, muscle-system processes, response to wounding, chemotaxis, regulation of hormone levels, muscle contraction, membrane potential, actin filament processes, vasculature development and angiogenesis. In bulk tissue, such terms may partly reflect differences in stromal, vascular or other non-tumour cell populations.

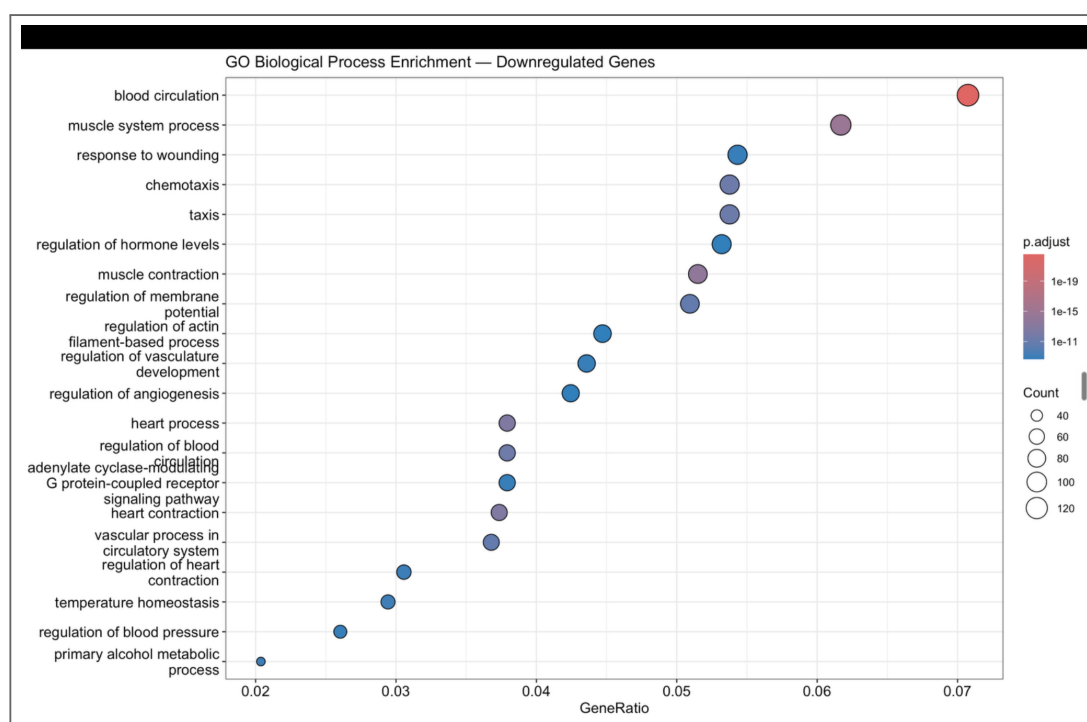


Figure 7. GO Biological Process enrichment of downregulated genes.

11. Discussion

The analysis demonstrates extensive transcriptional differences between primary breast tumour and normal breast tissue. The large number of significant DEGs, PCA separation and structured heatmap patterns support tissue-associated molecular remodeling. Enrichment of chromosome segregation, nuclear division and nucleosome-related processes among upregulated genes provides a coherent biological interpretation of increased proliferative and chromosome-maintenance activity.

The selected-gene analysis recovered significant expression differences for BRCA1, BRCA2, ESR1 and ERBB2. TP53 demonstrates the distinction between statistical significance and effect-size significance: its adjusted p-value was significant, but its fold change did not satisfy the predefined DEG threshold.

12. Limitations

- The sample groups are strongly unbalanced (1,111 tumour versus 113 normal).
- The model did not explicitly adjust for batch or centre effects.
- The comparison was analysed as an unpaired group analysis rather than a patient-paired model.
- Bulk RNA-seq combines tumour-cell and non-tumour-cell signals.
- GO enrichment is exploratory; a refined analysis should explicitly specify an appropriate gene universe/background.
- Mutation status, survival, treatment response and causal mechanisms were not analysed.

13. Conclusion

A complete TCGA-BRCA RNA-seq differential-expression workflow was implemented in R. The analysis identified 12,304 significant DEGs, including 8,186 upregulated and 4,118 downregulated genes. PCA, volcano plots and heatmaps provided complementary visual evidence, while GO enrichment connected DEG lists with biological processes. The project provides a reproducible foundation for future extensions such as survival analysis, molecular-subtype comparison, pathway analysis and protein-protein interaction analysis.

14. Reproducibility and GitHub

The GitHub repository is intended to contain the R analysis script, figures, result tables, README and this report. Raw TCGA data are intentionally excluded because of their large size; they can be retrieved from the Genomic Data Commons using the documented TCGAbiolinks query.

15. References

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3. Colaprico A, Silva TC, Olsen C, et al. TCGAbiolinks: an R/Bioconductor package for integrative analysis of TCGA data. *Nucleic Acids Research*. 2016;44:e71.
4. Yu G, Wang LG, Han Y, He QY. clusterProfiler: an R package for comparing biological themes among gene clusters. *OMICS*. 2012;16:284–287.

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6. Subramanian A, Tamayo P, Mootha VK, et al. Gene set enrichment analysis: a knowledge-based approach for interpreting genome-wide expression profiles. *PNAS*. 2005;102:15545–15550.